



Modelling route choice in public transport with deep learning

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Accepted: 16 February 2025
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Abstract

For choice problems in transportation, machine learning and deep learning are alternative methods to traditional choice models. While several works explored the potential of this technology for modelling mode choice, lower attention is given to route choice, especially in public transport. In this work, we propose a deep learning model designed specifically for route choice in public transport. The model can estimate a nonlinear utility function, allowing complex interactions among the variables; it can easily include non-alternative specific variables, such as weather or socio-demographic information. Moreover, compared to the traditional choice models, it numerically outperforms the Path Size Logit Model in prediction performance, and does not require pre-specification of the model by an experienced human modeler. These properties are particularly useful for route choice analyses, to capture possible heterogeneities or complex behavior, which are difficult to model a priori. We evaluated the interpretability of the model observing the marginal rates of substitution and applying Accumulated Local Effects, showing meaningful effects of the variables on the probability to choose an alternative. We tested the proposed model on a large-scale dataset based on GPS tracking. We considered both synthetic choices, to demonstrate the model properties, and real choices, to evaluate the model in practice. The results showed moderately better performance of the deep learning model compared to the Path Size Logit, confirming the possibility of using it for modeling and predicting route choice.

Keywords Deep learning · Route choice · Choice models · Machine learning · Public transport

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Introduction

Route choice is an important aspect of travel behavior, widely analyzed in literature given its several applications. In fact, information on travelers' preferences can be used for predicting traffic flows, or also for route recommendation. For public transport, modeling route choice is particularly important, since it helps design the public transport service and adapt it in case of disturbances in the network.

The most used approach in literature for route choice modeling combines a choice set (CS) generation algorithm with a route choice model (Anderson et al. 2017; Bovy 2009; Marra and Corman 2020; Yap and Cats 2021). The first one is a heuristic method, generating the available alternatives for a passenger to reach a destination. The second one is a model estimated from observed trips, estimating the probability of choosing each alternative in the CS. A route choice model is used both for route choice prediction and for understanding the choice process behind a certain choice. The most used models are discrete choice models. In particular, the Path Size Logit (PSL) and the Mixed PSL are the state of the art models for route choice in public transport (Marra and Corman 2020; Nielsen et al. 2021; Yap and Cats 2021). However, these models have limitations, such as the need of specifying a priori the structure of the utility function.

A recent stream of research in transportation is experimenting machine learning and deep learning as alternative to discrete choice models (Hillel et al. 2021; Wang et al. 2020a). Machine learning overcomes some of the limitations of discrete choice models and shows higher prediction performances, often at the cost of a lower interpretability (Sifringer et al. 2020). Most of these works focuses on mode choice (Buijs et al. 2021; Han et al. 2022; Lee et al. 2018; Sifringer et al. 2020) or drivers' route choice (Lai et al. 2019; Sun and Park 2017). To the best of our knowledge, there is no work applying deep learning for route choice in public transport.

This paper proposes a deep learning model for route choice in public transport. The model has higher prediction performance than traditional choice models, and does not need the preliminary specification of the choice model. The deep learning model allows complex interactions among variables (also known as features) and the inclusion of non-alternative specific variables. Therefore, such a model can capture choices based on complex utilities and identify possible heterogeneities. In practice, we design a deep neural network specifically for the prediction of route choice in public transport. As shown by Wang et al. (2020a), networks designed specifically for a certain task may have higher performance than fully connected networks. In fact, introducing expert knowledge in the network structure reduces the complexity of the model and the likelihood of overfitting. A key property of the proposed model is the permutation invariance, i.e. the output is independent on the ordering of the alternatives in input. This property is necessary for route choice, since a model must give the same result, no matter in which order the alternatives are considered. For example, the model must predict the same choice probabilities if we ask to compare the routes A, B and C or the routes C, B and A. In general, neural networks do not satisfy this property. In contrast, for mode choice, a deep learning model can depend on the ordering of the alternatives in input (e.g. first the variables of the bus, then the car, then the bike, etc...). This property makes a deep learning model designed for mode choice not suitable for route choice, and vice versa. Therefore, there is need of a specific deep learning model for route choice in public transport, not yet available in literature.

The proposed model has the following contributions:

- It is the first deep learning model in literature for route choice in schedule-based public transport. The model is permutation-invariant with respect to the ordering of the alternatives.
- The model can estimate a nonlinear utility function, allowing complex interactions among the variables.
- The model can include both alternative specific variables (e.g. travel time, number of transfers) and non-alternative specific ones (e.g. weather, socio-demographics).
- The model shows superior prediction performance than PSL, the state of the art model for route choice, allowing the estimation of a more complex utility function without the need of pre-specifying it as in traditional logit models, guaranteeing equal or higher performance with a proper tuning of hyperparameters.
- The model shows realistic interpretability metrics, similar to traditional choice models.

We compare the proposed model with the PSL on a large-scale dataset based on GPS tracking. We analyze both synthetic choices, to validate the properties of the model, and real choices, to evaluate the performance of the model in practice. Finally, we analyze the interpretability of the model, computing the marginal rates of substitution and applying Accumulated Local Effects, a technique describing how each feature affects the probability of choosing a route.

This paper is organized as follows: Section “[Literature review](#)” presents the literature; Section “[Methodology](#)” the methodology; Section “[Dataset and scenarios](#)” the dataset; Section “[Results](#)” the results; Section “[Interpretability](#)” investigates the interpretability of the model; Section “[Conclusions](#)” shows the conclusions.

Literature review

Route choice modelling

Route choice is a well-studied topic both in public and private transportation, given its several practical applications, such as for demand forecasting or service planning. A standard approach to study and model travel choices is discrete choice analysis, using probabilistic models for parameter estimation (Buijs et al. 2021; Zhao et al. 2020). The most known model is the Logit model, which determines the probability of choosing each alternative, based on a utility function, typically linear in the observed characteristics of the available alternatives. In route choice modelling, the PSL introduced by Ben-Akiva and Ramming (1998), Ben-Akiva and Bierlaire (1999) is a more advanced variant of the Logit, introducing a non-linear correction term for overlapping paths in the utility function. Recent works using PSL tackle for instance overlapping routes in road networks based on GPS data (Montini et al. 2017), and extend the PSL approach for public transport networks, where overlapping lines are a common issue (Tan et al. 2015). A more complex model considering also user heterogeneity is the Mixed PSL (Marra and Corman 2020; Nielsen et al. 2021). The PSL approach allows more complex model specifications and interaction terms which go beyond the linear-in-parameter model. Nevertheless, the recent papers addressing route

choice in schedule-based public transport networks, based on revealed data consider the utility specification as linear (Anderson et al. 2017; Marra and Corman 2020; Nielsen et al. 2021), with the exception of a piecewise linear (describing a critical crowding density) modelled in Yap and Cats (2021).

In recent years, a novel stream of research in transport-related choice problems experimented models based on machine learning and deep learning, as alternative to discrete choice models (Wang et al. 2020a; Zhao et al. 2020). Discrete choice models have several advantages, such as an established theory behind and a high interpretability of the models. Nevertheless, they also have limitations in the definition of the utility function and the practicality to understand the most suitable interactions among the observed variables. In contrast, machine learning models can directly identify non-linear relationships among the variables and allow the estimation of complex classification models, at the cost of a lower interpretability (Lee et al. 2018). To tackle this problem, the Logit can capture the interaction between correlated variables by adding interaction parameters, piecewise approximations, and transformations of inputs. However, considering all possible interactions becomes impossible with the increase of the number of variables (Kim 2021). As a result, the typical PSL used in state of the art research in public transport route choice based on revealed preference data effectively considers a utility function as a linear combination of the observed variables.

As highlighted by Hillel et al. (2021), a major difference is that the utility in discrete choice models is defined by the modeler prior to fitting the model, allowing the incorporation of established behavioral theory. According to Sifringer et al. (2020), this is the main limitation of those models, since the model specification is assumed known a priori, while in practice it is hard to determine the right utility specification. In contrast, machine learning models identify the relationship between input features and the class labels directly from the data, without a predefined utility. This allows identifying more complex relationships, which may lead to higher performances. In fact, several recent studies verified machine learning can outperform logit models in prediction accuracy (Zhao et al. 2020).

While several works applied machine learning for mode choice (see Section “[Deep learning for mode choice](#)”), there are very few works focusing on route choice. Moreover, they only focus on car traffic, and not on public transport, which has different characteristics, such as the availability of transfers, different modes, and routes with shared lines with different schedule.

The model specification and explanatory variables are substantially different, as a public transport network, considered in a schedule-based detail, requires the choice of a starting stop, waiting for a scheduled service, the possibility to have transfers. None of those aspects is present in road route choice models. As extensively argued in Anderson et al. (2017), studies using RP data for public transport route choice tackle very specific challenges which are non-existing in the wider literature of route choice modeling for road networks. Road networks can be studied by just looking at the space extent, and the interconnection of links, because a road link is always available for travelling. Public transport networks are more complex, and require both space and time extent modelled, because a public transport link describes the fact that the vehicle departs only at specific times. Just a handful of studies tackle real public transport networks where connectivity is pervasive, consider revealed preference data, with large sets of competitive different alternatives, identified by means of a choice set generation, including also different boarding stops and departure times.

For instance, Lai et al. (2019) compared different machine learning models with standard choice models for drivers' route choice, showing also the former ones can be adopted for behavioral analysis. Lee et al. (2005) compared an approach based on Neural Network and genetic algorithm with the logit model for drivers' route choice. They showed the Neural Network approach had better performance, given its ability to identify non-linearity in the decision-making process. Sun and Park (2017) compared Support Vector Machine and Neural Network for drivers' route choice modeling, based on a stated preference survey. The two methods had similar performance, despite a relatively simple Neural Network was used (one hidden layer) and the experiment was based on binary choices. Yao and Bekhor (2020) combined machine learning and discrete choice models for route choice modelling for drivers. However, their focus was the CS generation and the specification of the utility, while they estimated a standard Logit. Zhao and Liang (2023) applied deep inverse reinforcement learning for route choice in cars. Their model outperformed in prediction accuracy different baselines: (linear) PSL, Recursive Logit, and a fully connected neural network. However, their model required significant computation time, despite it was tested on a small network (712 links). Therefore, the model can be hardly applied in a schedule-based public transport network, like the one tested in this study, formed by 10 million links.

To the best of our knowledge, this is the first work designing a deep learning model for route choice in public transport. Therefore, the main contribution is the proposal of an alternative model for route choice in public transport, based on a neural network, specifically designed for this problem. The proposed model allows complex interactions among the variables, the estimation of a non-linear utility function, and the inclusion of non-alternative specific variables in the utility, without the need to prespecify any of those complex expressions. These properties allow the model to explain complex route choices in public transport, with a higher numerical performance than possible with traditional choice models.

Deep learning for mode choice

In this Section, we report the most recent works applying deep learning for mode choice. This is different from route choice, our focus, where only one work was identified (Zhao and Liang 2023), applying deep inverse reinforcement learning. Although the two problems are intrinsically different, they are both tackled applying discrete choice theory, but tailoring the model to the specific problem and using different features. Therefore, the literature of deep learning for mode choice can be useful to design a model for route choice.

Given the novel stream of research, the works in literature are experimenting different deep learning models or network structures for mode choice. Most of these models claim better prediction performance than the logit, while possibly trying to keep a certain degree of interpretability. Lee et al. (2018) compared four types of neural networks with the logit model for mode choice, showing higher prediction accuracy for the first ones. Zhao et al. (2020) compared seven machine learning models with the logit and the mixed logit on stated preference data for mode choice. The random forest had significantly higher prediction accuracy than the logit models.

Several works used generic neural networks, such as fully connected networks, widely applied in other classification problems (Buijs et al. 2021; Omrani 2015; Zhao et al. 2020). In contrast, Wang et al. (2020a) adapted the architecture of the network, introducing some knowledge from the field of mode choice. They showed a network based on underlying

assumptions of the random utility theory might lead to better results than a generic fully connected network. Siffringer et al. (2020) defined a deep neural network, which enhances the logit model with a novel utility specification. Namely, the utility is formed by a linear component (as in the logit) and a data-driven term derived from a fully connected network. The two terms are added together to form the systematic part of the utility. Their model outperformed traditional choice models, being also able to keep the interpretability of key parameters. A major limitation is that the sets of features used in the two components are mutually exclusive, which is too restrictive, since features used in the data-driven term may affect parameters in the linear component (Han et al. 2022). Similarly, Han et al. (2022) affirmed that incorporating expert knowledge into a neural network through special structure and constraints can improve the performance. Based on that, they propose a deep learning model outperforming the logit in prediction accuracy, while maintaining interpretability. Finally, Wong and Farooq (2021), Wang et al. (2021) applied a residual neural network, integrating it with the logit model in different ways.

Since the stream of research is quite recent, the literature still focuses on proposing new models, rather than establishing a single one to use. Different works tackled different aspects of choice modeling (e.g. nonlinear utility specification, interpretability, model fusion, prediction accuracy), and it is not possible to determine a model performing best a priori. Moreover, these models are tested mainly for mode choice, making not clear their applicability and performance for different choice problems, such as for route choice. Our work follows the same direction of most of the cited works, but for route choice, i.e. proposing a deep learning model, with an architecture tailored for route choice in public transport.

Interpretability of models

The high prediction accuracy of deep learning models comes at the cost of a lower interpretability (Alwosheel et al. 2021). To fill this gap, several works studied the interpretability of these models in transportation, especially in mode choice.

Some works designed the structure of the neural network to be interpretable to some extent (Arkoudi et al. 2021; Siffringer et al. 2020). For instance, Siffringer et al. (2020) proposed a network modelling the systematic part of the utility with two components: a linear component (such as in the Logit), interpretable; and a data-driven component, where no a priori relationship is assumed. Similarly, but with a different structure, Wong and Farooq (2021) proposed a model named ResLogit, specifying the utility function as sum of the deterministic component of observed characteristics and a neural network component, capturing the unobserved heterogeneity. Han et al. (2022) proposed a hybrid model combining a feed-forward neural network, learning taste parameters from individual characteristics, and a MNL model with manually specified utility.

Several techniques are used in choice problems to interpret machine learning models. Variable importance measures the importance of each feature for the prediction. In mode choice, Zhao et al. (2020) measures the importance of features using standardized Beta coefficients in a logit model and Gini index for random forest. Permutation variable importance permutes the values of a feature across the dataset, to see the decrease of model performance (Kim 2021). Unfortunately, most variable importance metrics provides only a ranking of features without information on the direction of influence of the feature to the outcome. In this regard, Partial Dependence Plots (PDP) show the effect of changing the value of a fea-

ture on the prediction, with magnitude and direction (Zhao et al. 2020). A main issue of PDP is the assumption of the features being uncorrelated. When the features are correlated (such as travel time and transfer time in route choice), changing the value of a single feature may lead to unrealistic data, which can bias the estimated effect of the feature. A similar method, which addresses the correlation issue, is Accumulated Local effect (ALE) (Kim 2021). An ALE plot shows the main effect of considering different values of a certain feature compared to the average prediction of the data.

SHapley Additive ExPlanation (SHAP) is a machine learning framework, describing the contribution of each feature for a single prediction. Zhao and Liang (2023) applied SHAP to understand the effect of the input features on the estimated reward in their Inverse Reinforcement Learning model for route choice. Unfortunately, SHAP may suffer of the same problem of PDP in case of correlated features, which makes it unsuitable for our study.

A method specific for neural networks is Layer-wise Relevance Propagation (LRP, Alwosheel et al. 2021), applied often in computer vision, leveraging the structure of a neural network and the estimated parameters to determine the contribution of each independent feature for a specific prediction. Therefore, LRP is used for local interpretability, i.e. a single prediction, and not for global interpretability, i.e. how the model works or a feature affects any prediction in general.

In standard discrete choice models, the interpretability of the model is examined in terms of behavioral outputs (e.g. marginal effects, elasticities, willingness-to-pay). These measures are hard to compute in machine learning models and are rarely examined (Zhao et al. 2020). However, (Wang et al. 2020b) showed deep neural networks can provide economic information as complete as classical discrete choice models, including choice probabilities, elasticities and marginal rates of substitutions. The economic information may be unreliable, with a smaller sample size or if considered for individual observations, while aggregated information is more reliable.

In the authors' opinion, the research on interpretability of deep learning models for travel behavior analysis is still at an early stage, often dependent or subordinate to the design of the model or the specific choice problem. In fact, novel models are often proposed (see Section "Deep learning for mode choice"), aiming at better performance than discrete choice models. The interpretability is studied in addition to that. Moreover, several techniques of explainable artificial intelligence have been applied, as alternative to traditional behavioral outputs, without a standard method valid for every model. For instance, the only work applying deep learning for route choice (Zhao and Liang 2023), used SHAP, which provides some degree of interpretability, despite less than traditional choice models. To the best of our knowledge, there are no studies on deep learning for route choice in public transport, and thus also on their interpretability.

Methodology

We propose a deep learning model for route choice in public transport. Our model builds on the PSL, allowing for more complex interactions among input variables. In fact, with a proper tuning of the neural network hyperparameters, the model has the same or better prediction performance of the PSL. Before describing the proposed deep learning model,

we briefly describe how the CSs are generated, the benchmark method (PSL), and how to represent the PSL with a neural network, since our model is based on that.

Choice set generation

In this section, we present briefly the CS generation algorithm used in this work, to generate the available alternatives for each observed trip in the dataset. The algorithm is described in detail in Marra and Corman (2020). Given an origin and a destination, the algorithm identifies a set of available routes using public transport. Each route is formed by walks to public transport stops, runs on public transport vehicles, and transfers. The algorithm has a coverage of 94% (observed trips identified in the CS) and is based on constrained enumeration, a family of deterministic methods. Namely, the algorithm identifies all available routes satisfying constraints on the travel time (maximum twice the shortest path) and the number of transfers (maximum 2). This last constraint is not particularly relevant, since in our test case there are very few trips with more than 2 transfers (see Table 1 for more details).

The algorithm can identify CSs up to 100 alternatives. In this work, we test CSs with different sizes. The CSs are sorted by travel time, including a transfer penalty of 5 min. The sorting does not affect the model estimation, but only the selection of the alternatives when smaller CSs are considered. This affects only marginally the results, since most of the users (87%) choose among the 5 shortest alternatives.

We consider the route chosen by a user matching the route in a CS if the same public transport lines are used (taking the same line at a near stop is considered as the same choice). Additional lines replaceable by short walks (< 700 m) are not considered (see discussion on walking distance in Marra and Corman 2020). The CSs considered in this work are generated from the timetable. Therefore, no information on delays or other disturbances are considered.

In Marra and Corman (2020) and in following works (Marra 2021; Marra et al. 2022) the algorithm was already applied to estimate a route choice model (PSL and Mixed PSL). Moreover, the model was tested on the same dataset of this work, estimating meaningful parameters, in line with the values in literature. We refer to Marra and Corman (2020) for more details on the CS generation algorithm, its limitations, and the estimation of traditional route choice models.

Table 1 Information on the tracking dataset

Period	03/04/2019–02/06/2019
Participants	172
Average days per person	22
Age	<18: 0%; 18–24: 33%; 25–34: 27%; 35–44: 21%; 45–54: 18%; 54+: 1%
Gender	Female: 43%; Male: 57%
Mode share	P.t.: 38%; Walk: 23%; Car: 15%; Bike: 13%; Mixed: 10%
Public transport trips in Zürich	2901
# transfers per trip	0: 60%; 1: 29%; 2: 9%; 3+: 2%
Public transport mode share	Tram: 52%; Bus: 38%; Train: 10%
Avg. duration per trip	21.7 min
Avg. air distance per trip	2.88 km

An alternative to generate a CS is to apply a link-based route choice model, like the recursive Logit. Despite link-based approaches not requiring assumptions on the choice set, they suffer for high computational costs, which prohibit their usage in large schedule-based networks, like the one of this study (10 million arcs). In fact, Meyer de Freitas et al. (2019) needed to simplify the public transport network of Zurich into a static network (i.e., removing all schedule information), to train the recursive logit. For a small network, Zhao and Liang (2023) show a much higher computation time for model optimization of link-based models compared to path-based ones, whose cost depends mainly on choice set generation (few seconds for our algorithm). Moreover, in their test-case, the PSL outperformed the recursive Logit, showing there is not a clear preference between the two models.

Path size logit

The PSL is the most used model for route choice in public transport (Nielsen et al. 2021; Yap and Cats 2021). This model was also estimated in Marra and Corman (2020), using the same dataset of this work. We here consider the typical PSL in the same linear-in-parameter form as those works in the state of the art; we are aware than in theory more complex model specifications of PSL are possible.

The PSL is an extension of the Logit, including an additional parameter in the utility (*PathSize*), to deal with correlated alternatives in the CS (i.e. routes with shared lines). In this work, the utility function (Eq. 1) includes the travel time in tram, bus, train and walking, the transfer time, the transfer penalty and the *PathSize* (Eq. 3).

$$U_{i,n} = V_{i,n} + \varepsilon_{i,n} \quad (1)$$

$$V_{i,n} = \beta_{tram} * tramTime_{i,n} + \beta_{bus} * busTime_{i,n} + \beta_{train} * trainTime_{i,n} + \beta_{walk} * walkTime_{i,n} + \beta_{tt} * transferTime_{i,n} + \beta_{transfer} * transfers_{i,n} + \beta_{PS} * PathSize_{i,n} \quad (2)$$

$$PathSize_{i,n} = - \sum_{line \in i} \frac{duration(l,n)}{duration(i,n)} \ln (times\ l\ occurs\ in\ CS) \quad (3)$$

The *PathSize* is defined as in Marra and Corman (2020), which is based on Bovy et al. (2008). An alternative i for sample n has a penalty (the $PathSize_{i,n}$) if it has a line in common (same public transport line l) with other alternatives considered in the CS of sample n . The penalty increases with the number of alternatives in the CS, having the line in common.

We consider also additional factors in increasingly complex PSL variants, which consider interaction terms. We considered the following 15 interaction terms related to the socioeconomic characteristics of the traveler of sample n : gender with all time-related variables describing alternative i :

$$(tramTime_{i,n}, busTime_{i,n}, trainTime_{i,n}, walkTime_{i,n}, transferTime_{i,n});$$

and age (three categories: ≤ 35 , $35-65$, ≥ 65) again with all time-related variables describing alternative i . Only for the group scenario, we considered also the interaction with temperature (to match the synthetic data we generated). We also considered the alternative i (or all

of them, in case more than one) which happens to correspond to the shortest path (shortest duration), interacted with gender. This resulted in an additional term.

As well as in the Multinomial Logit, the probability of choosing an alternative in the CS is the following:

$$p_{i,n} = P(i, n | \text{choice set}_n) = \frac{e^{V_{i,n}}}{\sum_{j,n \in \text{choice set}_n} e^{V_{j,n}}} \quad (4)$$

The PSL can be estimated maximizing the following loglog likelihood function:

$$L_{log} = \sum_{n=1}^N \sum_{j=1}^K y_{j,n} \ln p_{j,n} \quad (5)$$

with N =number of samples; K =number of alternatives; $y_{j,n} = 1$ if sample n has chosen alternative j , 0 otherwise; $p_{j,n}$ = probability sample n has chosen j .

An advanced variation of the PSL, named Mixed PSL, allows for user heterogeneity in the model. We do not consider this model since, in Marra and Corman (2020), on the same dataset of this work, it had similar performance to the PSL (i.e., no remarkable heterogeneity was identified).

Path size logit with a neural network

In this Section, we show how the Multinomial Logit can be represented and trained with a neural network, as already proven in previous works (Sifringer et al. 2020). Given that the PSL differs from the Multinomial Logit only by an additional parameter in the utility, the same equivalence is valid also for the PSL. This explanation eases the understanding of the novel model proposed in this work, with a more complex structure and described in Section “Deep learning model for route choice”.

Figure 1 shows a convolutional neural network equivalent to a linear Logit. In input there are the features representing each alternative (three in the Figure), such as the travel time in tram and the others in Eq. (2). In the next layer, for each alternative a dot product (shown

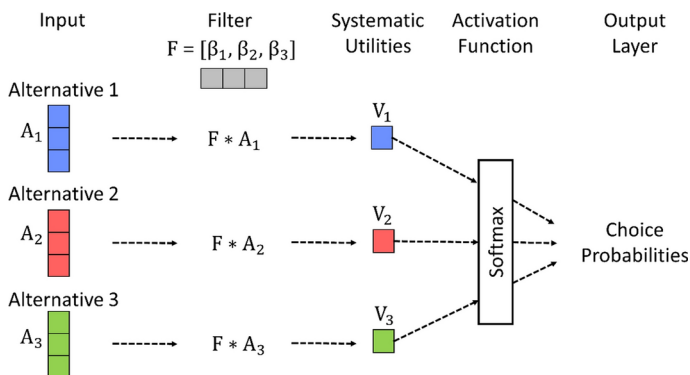


Fig. 1 Multinomial Logit represented with a convolutional neural network, considering 3 alternatives

in figure by *) is computed between the features of the alternative and a filter, containing the β s to estimate. The result of the dot product is the systematic utility in Eq. (2). Finally, a Softmax layer computes the probability to choose each alternative, as in Eq. (4). Estimating this network means estimating the β s in the filter. The network can be estimated minimizing the categorical cross-entropy, which is the following function:

$$H(p, y) = - \sum_{j=1}^K y_j \log p_j \quad (6)$$

with K =number of alternatives; p_j = probability to choose the alternative j (which is the output of the j -th neuron); $y_j = 1$ if j is the target class, 0 otherwise.

As also shown in Siffringer et al. (2020), minimizing the categorical cross-entropy (Eq. 6) in the neural network is equivalent to maximizing the Log likelihood (Eq. 5) in the Logit.

As a practical consideration, unavailable alternatives can be modelled including an additional variable in input for each alternative. This variable has value 0 if the alternative is available, while ∞ if unavailable (practically a big number, of a higher order than the utility). The corresponding β can be fixed to -1 . This variable will act as a penalty to the utility in case the alternative is unavailable, resulting in a probability to choose that alternative ≈ 0 .

Finally, we remark that in Fig. 1 only 3 alternatives are shown. The network comprises a single filter layer, which corresponds to linear-in-parameter model specifications. In this case, increasing the number of alternatives available does not increase the number of parameters to estimate (the β s). This behavior is similar to Logit, but the opposite from other neural network architectures. For instance, in fully-connected NN, with multiple layers, the amount of parameters depends on the amount of layers, and the amount of alternatives.

Deep learning model for route choice

Model structure and properties

Figure 2 shows the proposed deep learning model for route choice in public transport, which we named “**DL-route**”. As in Fig. 1, a CS of 3 alternatives is shown. In the Input layer, there are the alternative-specific features for each alternative (the ones in Eq. 2), and the non-alternative-specific features, named “External features”, including socio-demographic and weather information. Each group of alternative-specific feature is concatenated with the external features, forming a vector of the same size for each alternative. Afterwards, L convolutional layers are applied. Each convolutional layer has M filters and an activation function α . α is applied to the dot product (shown in figure by *) of the filter F with the input features, plus a constant term γ . All F and γ in all layers are the parameters to estimate. The last convolutional layer has only one filter, returning one value, the systematic utility. Finally, a Softmax layer is applied to compute the probabilities, as in Fig. 1.

Unavailable alternatives can be modelled as described in Section “[Path size logit with a neural network](#)” for the Logit. Namely, a penalty feature can be added to the utility in the last layer if the alternative is not available. This is similar to Nam and Cho (2020), applying an “Availability of Alternatives Function” in the last layer.

DL-route has a similar structure to the PSL, but more complex. Both models start with a vector for each alternative in input, and end with the systematic utility and a Softmax layer.

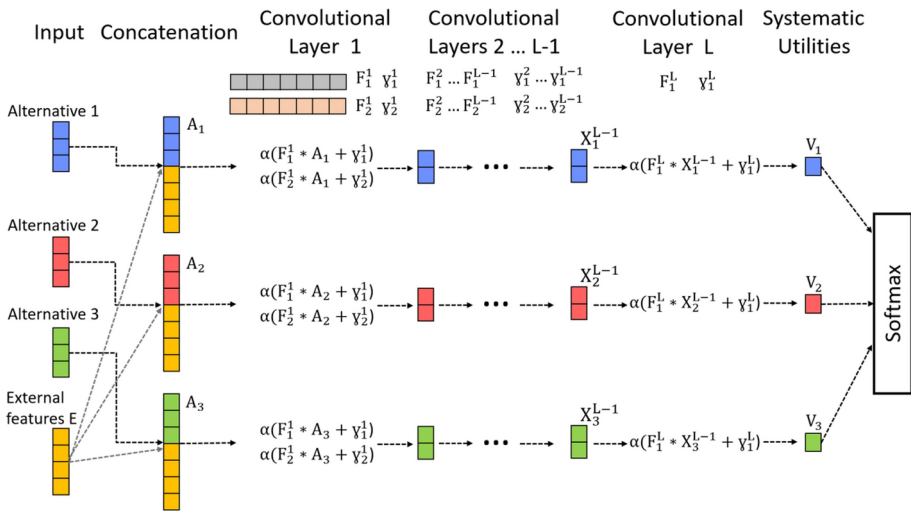


Fig. 2 Graphical representation of the proposed deep learning model. The example shows 3 alternatives, L layers and 2 filters per layer

The major differences are the inclusion of external features in input, the usage of multiple convolutional layers, and the activation function α , which, if non-linear, allows complex interactions among the features to compute the utility. The network of Fig. 1 corresponds to a typical linear PSL (matching the ones in the state of the art of our problem), expressing the linear utility by means of one layer and one filter. Instead, DL-route learns a non-linear utility applying multiple convolutional layers, with multiple filters and a non-linear activation function α . Therefore, with a proper tuning of the hyperparameters, DL-route guarantees similar or better prediction performance than the typical PSL. In fact, with $L=1$, $M=1$, $\alpha(x)=x$, and no external features, the model is equivalent to the typical PSL.

Since all the layers are convolutional, the same utility function is learned for each alternative. Moreover, including more alternatives in input (larger CS) does not increase the number of parameters to estimate. In fact, for $L \geq 2$, the number of parameters is the following:

$$\# \text{ parameters} = (F + E + 1) * M + (L - 2) * (M + 1) * M + M + 1 \quad (7)$$

with F and E , number of alternative-specific features and external features. Since the number of alternatives does not affect the complexity of the model (in terms of parameters to estimate), the model is scalable and can be used with CSs with different sizes.

In the general context of machine learning, Lee et al. (2019) affirms that models solving problems not depending on the order of the input elements should be permutation-invariant. Namely, in route choice, a model should predict the same probabilities, regardless of the order of the alternatives. This makes sense intuitively, since if we ask a model to compare two routes, A and B , we expect the same result, if it compares A with B or B with A . In general, neural networks are not permutation-invariant. Our model satisfies this property, since it does not depend on the order of the alternatives in input. In fact, regardless of the order, the same function is applied to each alternative, resulting in the same utility and choice probability. This property is extremely important for route choice modeling, where in general the

routes in the CS have not a predefined order. Not satisfying this property would make the model less robust to novel inputs and lead to interpretability problems. Moreover, the results would be unstable, depending on the order of the alternatives. In contrast, the alternatives in the mode choice problem can have assigned a fixed order (e.g. first the bus, second the car, third the bike). This difference reflects a different structure required for the neural networks for the two problems. For route choice, we propose a convolutional network, to guarantee the permutation-invariance property. For mode choice, most of the networks proposed in literature are not permutation-invariant (see Section “[Deep learning for mode choice](#)”). We remark that the proposed model is only one of the possible models that are permutation-invariant. For future research, we encourage the development of different models, not necessarily convolutional. For instance, parameter sharing techniques can be applied to models previously defined for other purposes, to make them permutation-invariant (since they are new models, the same properties and performance might not be guaranteed).

An alternative approach to a permutation-invariant network for route choice is predefining an ordering of the alternatives (e.g. the shortest one first). However, we discourage such approach, since the model would depend on an arbitrary criterion decided by the modeler. Finally, we remark that the model is permutation-invariant regarding the alternatives, but not regarding the features of each alternative, which must be provided in the defined order (i.e. in Fig. 2, the vectors of the three alternatives can be swapped, but each vector keeps the same ordering of features).

Summarizing, DL-route has by design the following features:

1. It can estimate a nonlinear utility function. Thanks to a nonlinear activation function α , complex interactions among the variables in input are allowed. This is similar to considering piecewise-linear terms and interaction terms in logit models.
2. It can include non-alternative specific variables in the utility. In literature, there are already models considering these variables, for instance including interaction terms in the utility (Menghini et al. 2010; Raveau et al. 2014).

And has the following advantages, compared to the linear PSL models found in the state of the art of the route choice modelling in public transport network:

3. The model guarantees similar or better prediction performance than the typical linear PSL, with a proper tuning of the parameters; and in general has higher numerical performance than traditional models.
4. Does not need a preliminary specification of the model, especially, no need to identify the type of interactions a priori; instead, DL-route directly learns complex interactions among all variables.

Model estimation, evaluation and benchmarks

We estimated the DL-route, following a standard machine learning approach. We divided the dataset (described in Section “[Dataset and scenarios](#)”) in 60% Training Set, 20% Validation Set and 20% Test Set. The model is trained on the Training Set and evaluated on the Validation Set, to identify the best set of hyperparameters. The hyperparameters tuned are: α (relu or tanh), L , M , **learning-rate**, **l2-regularization**. The performance of the model

are evaluated on the Test Set and shown in Section “[Results](#)”. We considered the following external features, including both socio-demographic and weather information: gender, a binary variable in our test case (0 or 1); income, represented with 4 categorical features (low, medium, high, not declared); age, humidity and temperature, modelled as numerical features, normalized to have 0 mean and unit standard deviation ($x' = (x - \mu)/\sigma$).

We compared the performance of the DL-route with two benchmarks, the typical PSL and a two-layer feed-forward network (2L-DNN). The structure of 2L-DNN is a classic fully-connected network: 1 input layer, 2 dense layers, a softmax layer. In this case, the amount of parameters increases with the amount of alternatives, which makes model estimation cumbersome. Number of nodes, activation function, l2-regularization and learning-rate are hyperparameters. These two models are the same benchmarks used by Zhao and Liang (2023), named PSL (linear) and DNN-PSL in their work. We do not compare with link-based models, like Zhao and Liang (2023) or the recursive logit, for computational infeasibility with our dataset, as explained in Section “[Choice set generation](#)”. Since the PSL does not have hyperparameters, both the Training Set and the Validation Set are used to train the PSL. We additionally consider variants of PSL with interaction terms. We evaluated the three models on the same Test Set in terms of **Log likelihood** (divided by the number of samples, $L'_{log} = L_{log}/N$) and **Prediction Accuracy** (#correct detection/#observations).

The software used for model estimation is keras (Chollet et al. 2015), using Adam as estimation algorithm, a batch size of 32, and the number of epochs determined with “Early Stopping”.

Interpretability

Marginal rates of substitution

Wang et al. (2020b) showed deep neural networks can provide economic information as complete as classical discrete choice models. We analyze the interpretability of DL-route, observing the choice probabilities and marginal rates of substitution (MRS). Given that all public transport trips in Zurich have the same monetary cost, we do not analyze the value of time, but how each route characteristic compares to one minute of travel time in tram (i.e. the MRS).

In a deep learning model, the probability derivative of alternative k with respect to $x_{j,n}$, with n sample and j feature, is the gradient of the choice probability function:

$$\partial p_k(x_n) / \partial x_{j,n} \quad (8)$$

The MRS is thus the following:

$$MRS(x_{j,n}, x_{m,n}) = - \frac{\partial p_k(x_n) / \partial x_{j,n}}{\partial p_k(x_n) / \partial x_{m,n}} \quad (9)$$

The MRS of the model can be computed averaging the MRS among all observations in the test set. In fact, despite DNN can suffer of local irregularities of probability derivatives, aggregate information (such as the average) are more reasonable (see Wang et al. 2020b). In the PSL, the MRS between two features is simply the ratio between the estimated β s in

Eq. (2). We compute the MRS only for the alternative-specific features, since for external features there is not a clear relationship between a feature and each alternative (e.g. the temperature may lead to a positive derivative for some alternatives, while negative for others, leading to a mean close to 0).

Accumulated local effect

To observe how the choice probability varies based on a certain feature, several methods have been applied in literature. As shown in Section “[Interpretability of models](#)”, most methods of explainable artificial intelligence (PDP, SHAP, or permutation feature importance) may lead to unrealistic results in case of correlated features. Similarly, Wang et al. (2020b), Siffringer et al. (2020) compute traditional economic information in DNN, such as substitution patterns between alternatives based on a certain feature. To do so, they hold all variables constant, while varying the selected feature. This may be problematic for correlated features, since it may lead to unrealistic instances. For example, holding the travel times constant to a small value, while increasing the transfers, may lead to unrealistic instances, never observed in the data or available in reality. A model applied on those instances may have an unexpected behavior. Therefore, in this work, we apply Accumulated Local effect (ALE), a method able to deal with correlated input features (Apley and Zhu 2016; Khoda Bakhshi and Ahmed 2021), which are common in route choice problems.

ALE is a method of explainable artificial intelligence, showing the main effect of a feature to the prediction (Apley and Zhu 2016). Practically, an ALE plot shows the effect of considering different values of a certain feature compared to the average prediction of the data. It can identify complex nonlinear relationships between the input variable and the prediction. In this work, we show how different values of input features (e.g. walking time) affect the choice probability of a certain route.

Here, we summarize how to compute the ALE curve, while we refer to the original paper (Apley and Zhu 2016) for a formal and detailed explanation of the method. Given a feature of interest j , the core idea is to divide the range of possible values of the feature into K intervals. Then, ALE evaluates how changing the value of the feature from one interval to the next one affects the prediction. The ALE can be computed as follows:

$$ALE_j(x) = \sum_{k=1}^{k_j(x)} \frac{1}{|N_j(k)|} \sum_{i: x_i \in N_j(k)} [f(z_{k,j}, x_{i,j}) - f(z_{k-1,j}, x_{i,j})] \quad (10)$$

To compute the ALE for feature j at value x , ALE identifies the interval where x lies among the K intervals, $k_j(x) = [z_{k-1,j}, z_{k,j}]$. Then, it identifies the set of observations in that interval, $N_j(k)$, i.e. each observation with feature $j \in k_j(x)$. For each observation $x_i \in N_j(k)$, it computes the predicting function f (the trained model), replacing the value of the feature j , with the upper bound and lower bound of the interval ($z_{k,j}$ and $z_{k-1,j}$). The values of the other features $x_{i,j}$ ($j \neq j$) do not change. The difference between the two predictions, averaged among all data points in the interval, gives the local effect of changing the feature j in that interval. This represents the change in prediction from $z_{k-1,j}$ to $z_{k,j}$. To know the change in prediction from $z_{0,j}$, the local effects are accumulated across all previous intervals (first sum, from 1 to $k_j(x)$).

The ALE is usually centered to have 0 mean, being interpreted as the effect of feature j on the prediction compared to the average prediction. A larger number of intervals K leads to a more detailed plot, at the cost of fewer data points per interval, which may lead to unreliable results. We remark that ALE plot requires an ordering of the possible values of a feature. Therefore, an ALE plot cannot be defined for categorical features, like gender (unless a similarity function is defined to compare categories). We computed the ALE plots using the python library *alibi* (Klaise et al. 2021).

An ALE plot can also consider two features instead of one, showing the additional effect of their interaction, without including the main effect of the two features. In this work, we do not show these second order effects, since from preliminary analyses we identified marginal effects on the probability (between 0 and 2%), compared to first order effects (up to 30%). Moreover, second order effects are more difficult to interpret, and less reliable, since each interval K is based on two features and therefore it includes fewer data points.

Dataset and scenarios

Tracking dataset

We tested the DL-route model on realized data, collected via GPS tracking. The dataset was collected in 2019, in Zürich (Switzerland), consisting of 2901 public transport trips of 172 users. The data were collected with a smartphone application, the *ETH-IVT Travel Diary*, which collected GPS data continuously for around 3 weeks (Marra et al. 2019). The GPS data have been processed with a mode detection algorithm, which automatically derives a travel diary for each user. Practically, first the activities and trips are identified; then, the stages composing each trip and the transport modes used. The algorithm exploits Automatic Vehicle Location data (AVL) of Zürich public transport network to identify also the public transport line and vehicle used. Information on lines and vehicles are particularly important to study route choice in public transport, as in this work, since knowing only the mode is not sufficient to know the route. The mode detection algorithm has an average accuracy of 86% and has already been validated in previous studies. In particular, Marra and Corman (2020) used the same dataset of this study, showing a realistic mode share and realistic estimations of route choice models. We refer to Marra et al. (2019) for more details on the smartphone application and the mode detection algorithm.

Table 1 shows general information on the dataset. Among all trips observed from the tracking, we consider only the public transport trips inside the city of Zürich, which are 2901. 40% of them includes at least one transfer, and the most used mode is the tram, followed by the bus and the train. A complete description of the socio-demographic characteristics and the representativeness of the dataset is available in Marra (2021), Chapter 5.4. Moreover, the same dataset was used in Marra and Corman (2020), to study route choices with traditional models, like the PSL, a benchmark in this work.

Choice scenarios

For each of the 2901 public transport trips, we determine a CS, containing the available alternatives. Each set of alternatives corresponds to one classification instance for our

model. The alternative in the CS matching the trip of the user has a target value equals to 1, the others 0. In this work, and as a standard process in comparison of route choice models (see Anderson et al 2017), we consider only trips where the choice of the user is included in the respective CS (2719 trips $\approx$ 94% of 2901). In fact, it is not obvious how to determine the values of some features for trips non-observed in the CS (e.g. travel time). In fact, while the CSs are based on the timetable, the observations are based on actual trips, during actual operations, which may differ from the planned ones. This affects the model estimation only limitedly, since only 6% of the trips are discarded.

Despite the available CSs being arbitrarily large (up to 100 alternatives), in this work we consider 5 alternatives per CS, including the chosen one and the 4 shortest alternatives, in terms of travel time, with a transfer penalty of 5 min (this penalty is used only to select the alternatives and it does not affect the model). In Section “Choice set size variation and computation time”, we repeat some of the analyses with larger CSs, showing the model can scale well with the number of alternatives, which does not affect the performance and the conclusions of this work. In fact, as shown in the Section “Results”, the number of parameters of the model does not depend on the CS size.

We evaluated the DL-route on three different scenarios. All scenarios are based on the same dataset (described in Section “Tracking dataset”), but they differ on the alternative chosen in the CS. Two are based on synthetic choices, to show the properties of the model and substantiate the second and third contribution identified in the introduction; and one on real choices, to observe its performance in practice. The three scenarios are the following:

1. Real scenario: the chosen alternative is the one observed from the tracking.
2. Nonlinear scenario: the chosen alternative is determined by a deterministic nonlinear utility function. This scenario proves that the DL-route can learn a nonlinear utility. We considered the following nonlinear utility (travel time in seconds):

$$U = -(transfers + 1)^2 * (busTime^2 + trainTime^3 + 10tramTime) * \sqrt{walkTime + 2 * transferTime} \quad (11)$$

3. Group-based scenario: the chosen alternative is determined by a deterministic linear utility (Eq. 2, time in seconds), which depends on the gender of the participant (0 or 1 in this text) and the atmospheric temperature ($^{\circ}\text{C}$). This scenario proves DL-route can include non-alternative specific variables in the utility. The values of the coefficients ($\beta_{tram}, \beta_{bus}, \beta_{train}, \beta_{walk}, \beta_{tt}, \beta_{transfer}, \beta_{PS}$) are the following:

- Gender=0, temperature ≥ 12 : -1, -1, -10, -0.1, -0.1, -300, -0.6
- Gender=0, temperature < 12 : -1, -1, -0.1, -10, -10, -900, -0.6
- Gender=1, temperature ≥ 12 : -1, -10, -0.1, -1, -1, -300, -0.6
- Gender=1, temperature < 12 : -0.1, -1, -0.1, -0.1, -10, -1800, -0.6

The utilities chosen in the nonlinear and group-based scenarios are purposefully arbitrary and do not have any particular meaning. We tested different utilities to verify the same properties. A rigorous demonstration that any function can be approximated by a CNN is out of the scope of this paper, and we refer to Zhou (2020). We did not choose realistic utili-

ties, to distinguish the objectives of scenario 2 and 3 (to show the properties of the model) from scenario 1 (to apply the model with actual route choice observations). Since in the synthetic scenarios variables do not have a specific meaning, we consider a PathSize=0 for the Nonlinear scenario (so that the effects of the other parameters are more evident); and PathSize=-0.6 for the Group-based scenario.

In all three scenarios, we report values of the DL-route model and for reference also the values estimated by a set of PSL models with different specifications, in terms of prediction accuracy and Log likelihood. In the real scenario we compare the numerical performance of DL-route with PSL and also with 2L-DNN. To evaluate the robustness of the DL-route model, we included a random perturbation in the two synthetic scenarios. Namely, we added to the utility a Gaussian noise with 0 mean and standard deviation σ , equal to 10% or 30% of the utility.

Results

Nonlinear scenario

In this section, we compare the DL-route with the PSL on the nonlinear scenario. The dataset contains 2719 observations, one for each trip collected. We considered CSs including the 5 shortest alternatives (as explained in Section “[Choice scenarios](#)”). The chosen alternative is the one maximizing the nonlinear utility in Eq. (9).

Table 2 shows the performance of the two models on the test set. Considering no noise in the utility ($\sigma=0$), the prediction accuracy of the DL-route is 98%. This shows the model is able to estimate almost perfectly the nonlinear utility function. In contrast, the typical PSL, which considers only a linear utility, has an accuracy of only 85%, and worse Log likelihood. Considering more complex PSL models improves marginally accuracy and Log likelihood.

Including noise in the utility worsens the performance of both models, as obvious, since a component of the utility cannot be predicted. With a Gaussian noise with $\mu=0$ and $\sigma=30\%$ of the utility, the prediction accuracy of the DL-route drops from 98 to 83%, while the one of the PSL drops from 85 to 76%. Therefore, DL-route has still better performance and can be applied also when the utility is not fully determined by the available features, but includes additional components, such as a random noise.

In this Section, we showed that the DL-route model is able to learn a nonlinear utility. However, a theoretical proof that the model can learn any nonlinear utility function is out of the scope of this paper. In internal tests (not shown in this paper), the model was able to estimate different nonlinear utilities. Considering a different CS size does not change the results in this Section. However, the CS size may affect the tuning of the neural network. Further considerations are in Section “[Choice set size variation and computation time](#)”.

Group-based scenario

In this Section, we analyze the performance of the DL-route on the more complex setup of the group-based scenario with both alternative-specific and non-alternative-specific terms. Again, we report the values of the PSL as a reference. As in the previous section, we consid-

Table 2 Performance comparison between DL-route and PSL on the nonlinear scenario

Nonlinear scenario	No noise	$\sigma = 10\%$	$\sigma = 30\%$
DL-route	98%	93%	83%
Prediction accuracy			
PSL	85%	81%	76%
Prediction accuracy			
PSL+interactions	85%	83%	76%
Prediction accuracy			
PSL+shortestpath	83%	82%	73%
Prediction accuracy			
PSL+interactions+shortestpath	83%	81%	75%
Prediction accuracy			
DL-route	-0.08	-0.20	-0.48
Log likelihood (L'_{log})			
PSL	-0.53	-0.57	-0.69
Log likelihood (L'_{log})			
PSL+interactions	-0.51	-0.51	-0.74
Log likelihood (L'_{log})			
PSL+shortestpath	-0.51	-0.50	-0.65
Log likelihood (L'_{log})			
PSL+interactions+shortestpath	-0.50	-0.50	-0.68
Log likelihood (L'_{log})			
Training set	1631		
Validation set	544		
Test set	544		
Interactions	Gender $\times$ Age; Shortestpath $\times$ Gender		

Different σ for a Gaussian noise are considered (10% and 30% of the utility)

ered CSs with the 5 shortest alternatives. The chosen alternative is the one maximizing the utility of the group of that user (see Section “[Choice scenarios](#)”). We considered gender and temperature as external parameters. Table 3 shows the performance of the two models on the test set. The DL-route has a prediction accuracy of 93%. This shows the model is able to include non-alternative specific variables, and to learn different utilities depending on user and weather’s characteristics. In contrast, the typical PSL has an accuracy of only 63%. In fact, the typical linear version of it cannot include non-alternative specific variables in the utility (adding such variables in the utility results in the same additional value to the utility of each alternative, which does not change the choice probability). The consideration of interaction terms describing possible roles of the non-external features (i.e. the more complex PSL models) improves accuracy and Log likelihood, with some erratic pattern across the different models, and at the cost of determining a suitable model specification (in this case we try to match at best the ground truth of the synthetic choices). In the specific test reported, we considered the simple interactions (gender, age), and the complex interactions (gender, age, temperature, matching the categories identified when generating the synthetic data), plus the possibility of having socioeconomic terms interacted with the shortest path alternative. The performance improve substantially from the simplest to the more complex model, but those latter are still worse the DL-route.

Including noise in the utility worsens the performance of both models, as in the Non-linear Scenario. With a Gaussian noise with $\mu=0$ and $\sigma=30\%$ of the utility, the prediction accuracy of the DL-route drops from 93 to 66%, while the one of the PSL drops from 65 to

Table 3 Performance comparison between DL-route and PSL on the Group-based Scenario

Group-based scenario	No noise	$\sigma = 10\%$	$\sigma = 30\%$
DL-route	93%	77%	66%
Prediction accuracy			
PSL	65%	53%	47%
Prediction accuracy			
PSL+ <i>simple_interactions</i>	73%	62%	50%
Prediction accuracy			
PSL+shortestpath	67%	57%	54%
Prediction accuracy			
PSL+ <i>simple_interactions</i> +shortestpath	74%	63%	54%
Prediction accuracy			
PSL+ <i>all_interactions</i>	72%	62%	53%
Prediction accuracy			
PSL+ <i>all_interactions</i> +shortestpath	80%	68%	54%
Prediction accuracy			
DL-route	-0.1210	-0.1517	-0.925
Log likelihood (L'_{log})			
PSL	-1.078	-1.217	-1.271
Log likelihood (L'_{log})			
PSL+ <i>simple_interactions</i>	-0.924	-1.056	-1.209
Log likelihood (L'_{log})			
PSL+shortestpath	-0.984	-1.167	-1.201
Log likelihood (L'_{log})			
PSL+ <i>simple_interactions</i> +shortestpath	-0.845	-1.016	-1.142
Log likelihood (L'_{log})			
PSL+ <i>all_interactions</i>	-0.745	-0.983	-1.165
Log likelihood (L'_{log})			
PSL+ <i>all_interactions</i> +shortestpath	-0.587	-0.843	-1.120
Log likelihood (L'_{log})			
Training set	1631		
Validation set	544		
Test set	544		
Non-alternative specific variables in DL-route	Gender, Temperature		
<i>simple_interactions</i> terms in PSL	Gender, Age		
<i>all_interactions</i> terms in PSL	Gender, Age, Temperature		
Shortest path	Gender, Shortest path		

Different σ for a Gaussian noise are considered (10% and 30% of the utility)

47% (from 74 to 54% for the more complex PSL model). Therefore, DL-route has still better performance and can be applied also when the utility is not fully determined by the available features, but includes random noise.

In this section, we showed that the DL-route is able to learn a utility depending on non-alternative specific features. However, we evaluated only one way of considering these features in the utility, i.e. having a different utility for 4 different cases. For instance, a different scenario may consider interaction terms (e.g. *temperature*walkTime*) in the utility, combining non-alternative and alternative specific variables. Considering all possible interactions

among variables in synthetic scenarios is out of the scope of this paper. Moreover, this section does not focus on studying taste-heterogeneity among users; thus, we refer to Section “[Marginal rates of substitution](#)” for comments on the heterogeneity in the real scenario, and a possible comparison with the Mixed PSL.

Real scenario

Table 4 compares the DL-route with the PSL and 2L-DNN on the real scenario. The CSs considered include the chosen alternative and the 4 shortest ones, in terms of travel time with a transfer penalty of 5 min (5 in total, as explained in Section “[Choice scenarios](#)”). We estimated the DL-route with and without external features. The DL-route has higher prediction accuracy and better Log likelihood than any model based on PSL. Therefore, the DL-route has better performance than the current state-of-the-art model in route choice in public transport. This confirms DL-route is a new alternative model for route choice prediction in public transport.

For PSL, we considered also the three variants with interaction terms (gender, temperature) with time0-related variables; and with the interaction between shortest path and gender; and with both, as in the previous sections. The performance of more complex PSL forms remain stable or get very slightly worse for what concerns accuracy, and get very slightly better for what concerns Log likelihood.

However, despite the DL-route having better performance, the prediction accuracy is not particularly higher (76% against 73%). This shows that a linear utility (PSL), without external features, is sufficient to predict correctly 73% of the choices. The best model across those considering interaction terms allows an increase to 75% accuracy. Considering the most complex and best performing model, the 24% of incorrect predictions may be choices not explainable with the features available of the users and the alternatives (e.g. they may depend on vehicle crowding, or specific preferences of the user).

Comparing the DL-route model with and without external features, the performance is similar. This shows considering the chosen variables (listed in Table 4) in the DL-route model does not improve significantly the performance. Therefore, weather information and socio-demographic information have marginal effects on route choice, i.e., based on the experiments, users base their decisions mainly on other factors, such as the alternative-specific variables. This is in accordance with Marra and Corman (2020), which estimated a Mixed PSL on the same dataset, showing low heterogeneity among the users in the cost perception.

An alternative explanation of the marginal increase of performance of including external features may be the overfitting, since including additional features in the model increases the number of parameters to estimate. However, we exclude it as the main factor, since in our experiments, we tried different combinations of external features, tuning different values of the hyperparameters, and we did not identify any significant increase of performance.

We remark that the marginal effects of external features are related to our dataset in Zürich. Further experiments with different datasets are needed to generalize the outcome to the Swiss population, which is out of the scope of this paper. In fact, the low heterogeneity identified among the users may be partially due to the data collection or the CS generation algorithm.

Table 4 Performance comparison between DL-route, PSL and the 2L-DNN on the real scenario

Real scenario	DL-route (with external features)	DL-route (without external features)	PSL	PSL + inter action	PSL + shortest path	PSL + inter action + shortest path	2L-DNN
Prediction accuracy	76%	74%	73%	75%	72%	74%	65%
Log Likelihood (L'_{log})	-0.72	-0.73	-0.78	-0.78	-0.76	-0.76	-1.05
External features	Gender, Age, Income, Humidity, Temperature, Hour of the day	None	None	Gender, Age	Gender, Shortest path	Gender, Age, Shortest PATH	Gender, Age, Income, Humidity, Temperature, Hour of the day

The performance of 2L-DNN is much lower than the other two models. This is because a standard neural network is not suited for route choice, which requires permutation-invariance. Since the features of all alternatives may interact with each other, the number of parameters is much larger than for DL-route, and it explodes with the number of alternatives. This makes more difficult the training and tuning of the network.

Choice set size variation and computation time

In this Section, we evaluate again the DL-model on the Real Scenario (the one in Table 4, with external features), but considering different CS sizes. We want to evaluate the change in performance between considering few alternatives (simpler problem) and many alternatives (harder problem). We remark the complexity of the model does not depend on the number of alternatives considered, as shown in Eq. (7), since the same number of parameters is estimated.

Table 5 compares CS sizes of 5, 10 and 40 alternatives. A CS of X alternative includes the chosen alternative and the $X-1$ shortest ones. The prediction accuracy is comparable among the three cases, both for DL-route and PSL. In particular, the accuracy considering 40 alternatives (73%) is very close to considering 10 (74%). This confirms what the theory suggests: DL-route can consider CSs with different size and its performance do not depend significantly on it.

The main reason for similar accuracy with more alternatives is that most of the users choose the shortest ones. As explained in Marra and Corman (2020), 87% of times the observed trip is among the first 5 shortest alternatives and 90% of times among the first 10 (further information on CS coverage in Marra and Corman 2020). Hence, including more than 10 alternatives does not affect the model estimation significantly, since in most cases the additional alternatives are suboptimal and seldom chosen. Both DL-route and PSL predict a high probability for the shortest alternatives, while a very low probability for the suboptimal ones, which do not affect the prediction accuracy of the model. In theory, the performance may decrease with a larger CS, if many alternatives are similar to each other (which is quite unrealistic for our scenario).

The computation time of both models is low. On a general-purpose laptop, PSL in its simplest form required on average 0.27 s in a test-case with 5 alternatives, while 1.7 s with 20 alternatives. This time is particularly short since the model is linear and the input variables are only 7. DL-route took 20.7 s with 5 alternatives, while 29 s with 20 alternatives. The little difference between 5 and 20 alternatives is due to the convolutional nature of the network (i.e. the number of parameters does not depend on the number of alternatives). However, we performed cross-validation to select the best hyperparameters. Namely, we tested several DL-route with different configurations, and we selected the best one. We tested 100 configurations (total time of ≈ 35 min), even if a different number can be chosen based on the computational availability.

Table 5 Performance comparison of the DL-route and PSL on the real scenario, with different CS size

	CS size 5	CS size 10	CS size 40
DL-route	76%	74%	73%
Prediction accuracy			
PSL	73%	73%	72%
Prediction accuracy			
PSL+interactions	75%	72%	72%
Prediction accuracy			
PSL+shortestpath	72%	71%	71%
Prediction accuracy			
PSL+interactions+shortestpath	74%	71%	72%
Prediction accuracy			
DL-route	-0.72	-0.82	-0.91
Log likelihood (L'_{log})			
PSL	-0.78	-0.88	-0.94
Log likelihood (L'_{log})			
PSL+interactions	-0.78	-0.88	-0.93
Log likelihood (L'_{log})			
PSL+shortestpath	-0.76	-0.87	-0.94
Log likelihood (L'_{log})			
PSL+interactions+shortestpath	-0.76	-0.87	-0.93
Log likelihood (L'_{log})			

Table 6 Marginal Rates of Substitution w.r.t. travel time in tram (minutes)

	PSL	DL-route
tramTime	-1	-1
busTime	-1.12	-1.03
trainTime	-1.18	-1.22
walkTime	-2.58	-2.24
transferTime	-1.03	-1.04
Transfers	-14.9	-15.9
PathSize	70.2	18.0

CS size = 40

Interpretability

Marginal rates of substitution

Table 6 shows the MRSs for all features with respect to the travel time in tram. All values have negative sign (the PathSize is negative by definition), showing all derivatives have the same direction, which is realistic since all features penalize the utility. The magnitudes are coherent with previous studies on the same dataset (Marra and Corman 2020). For instance, the travel time in bus and train are perceived as 1.03 and 1.22 times more than in tram, while an additional transfer is perceived as a penalty of 15.9 min. Overall, the MRSs are very similar between DL-route and PSL. This shows DL-route interprets the relationships between features, coherently with traditional choice models. We refer to Marra and Corman (2020) for a detailed interpretation of those values (for PSL) and a comparison with the literature, showing they are realistic. The importance of the PathSize is discussed in the Appendix.

In this work, we do not consider the heterogeneity of the MRSs among users, given the limited number of trips per users in the test-set (5 on average). In fact, economic information

obtained from DNN are more reliable at the aggregate level, while unreliable at individual level, especially with a small sample size (Wang et al. 2020b). Moreover, estimating the Mixed PSL performed similarly to the PSL, identifying low heterogeneity in the current dataset (Marra and Corman 2020). Therefore, for a future work, we plan to study a long-term dataset, allowing the estimation of a different MRS for each user, and to compare DL-route with the Mixed PSL, to evaluate the heterogeneity of the interpreted parameters. This would be possible since larger sample sizes may reduce local irregularities and improve the interpretability at individual level (Wang et al. 2020b).

Accumulated local effects

Figure 3 shows the Accumulated Local Effects for the alternative-specific features for DL-route and PSL, with a CS of size 5. The plots show the change in probability of choosing an alternative, with different values of the feature of that alternative. The plots for the 5 alternatives are analogous, since the models are not influenced by the order. As an example of interpretation of ALE, an alternative with 0 transfers has $\approx 15\%$ higher probability than average to be chosen, while an alternative with 1 or 2 transfers have $\approx -10\%$ and -35% to be chosen.

Both for PSL and DL-route the probability is decreasing for higher values of the alternative-specific features. This is realistic since higher travel time and number of transfers decrease the appealing of a route. The absolute values, representing the changes in choice probability, are comparable between the two models. The differences in the last deciles of the plots (dotted in the figures) are less relevant, since very few data are available in those ranges. This shows that the two models interpret the features in a similar way. Therefore, DL-route is an explainable model, with realistic interpretation of the features, comparable to the PSL.

The range of probability-change is higher for the walking time and the transfers, showing these two features are the ones influencing the most the choice probability. This is in line with the theory and the MRS in Table 6, since walking time is generally perceived worse than travel time, and both models identified a transfer penalty of ≈ 15 min. Regarding the shape of the plot, the walking time has a sigmoid shape, as expected by the Logit. The bus and tram times have a more linear shape, which, given the smaller range of probability change compared to the walking time, can also be seen as the central part of a sigmoid. When analyzed by ALE, the train and transfer time have a small range of probability change. Given the dataset used which focuses on an urban situation, this is expected since only 10% of observed trips are done by trains.

The ALE plot for personal features is not shown, since for categorical variables (e.g. gender) there is no definition of ALE, while numerical features (e.g. temperature, age) have little changes of probabilities ($< 1\%$). The PathSize factor is also of limited importance ($< 0.02\%$).

The similar performance and interpretation of the two models show that a model based on deep learning is a valuable alternative to standard choice models for route choice. However, the similar performance can also be explained by the dataset itself, since 63% of the time the users choose the shortest alternative (Marra and Corman 2020), with low heterogeneity among them. Therefore, a linear utility is sufficient to explain around two thirds of the observed choices. This is one of the factors explaining why DL-route performs only a little better than the PSL. In contrast, in case of choices based on complex nonlinear relationship,

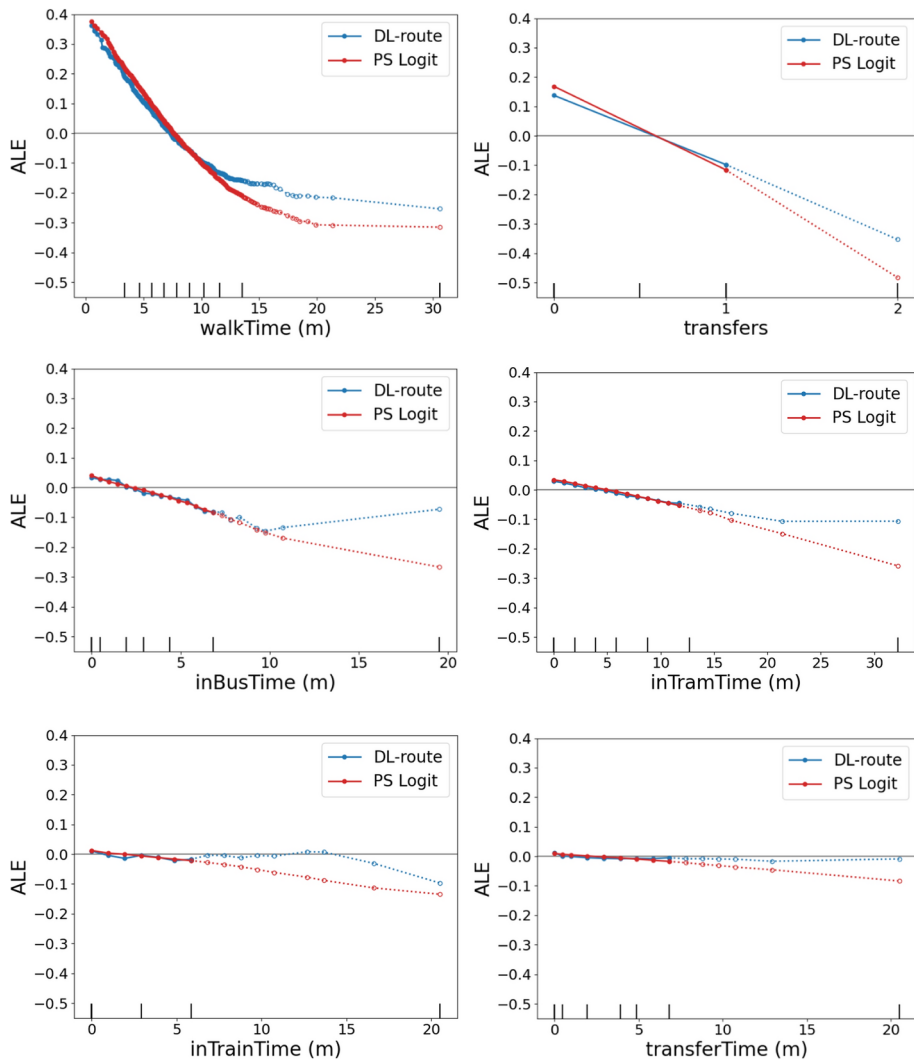


Fig. 3 ALE plots for DL-route and PSL. The plots show the change in probability of choosing an alternative, for different values of the features of that alternative. In black on the horizontal axis, there are the deciles of the feature. Last decile is dotted, as it is less reliable

as shown with the synthetic datasets, the performance of DL-route is significantly higher than the PSL, even when noise is present in the data.

One limitation of DL-route (and other machine learning models) compared to PSL is that it may be unstable to new inputs out of the distribution used for training the model. In fact, despite PSL and DL-route showing similar interpretation of parameters and choice probability, DL-route does not guarantee a specific output to a novel input. In contrast, PSL has a linear utility, clearly stating how the model will behave with a new input. This limitation can be mitigated using a large and representative dataset for training.

Conclusions

In this work, we proposed a novel method for predicting route choice in public transport, based on deep learning. The proposed model, named DL-route, is permutation-invariant with respect to the ordering of the alternatives. This characteristic is strongly desired for route choice, in contrast to mode choice, since the alternatives in input do not have a pre-defined ordering.

The model outperforms the PSL, providing two additional advantages. First, it is able to infer a nonlinear utility function. Second, it can include non-alternative specific variables, allowing complex interactions among all variables. We verified these two properties on synthetic data, showing also better performance than the PSL. We tested the proposed model on a real dataset, collected via GPS tracking. The model performed better than the PSL and a deep-learning-based benchmark (2L-DNN), showing its validity for route choice prediction.

DL-route can consider CSs with different size and its performance does not depend significantly on them. This property is guaranteed by the convolutional structure of the neural network, making the number of parameters independent on the number of alternatives.

We evaluated the interpretability of the two models computing marginal rates of substitutions and Accumulated Local Effects plots. The results showed similar interpretations of the alternative specific features in the two models, identifying the number of transfers and the walking time as the main features affecting the choice probability. Therefore, the interpretation of different values of each feature by DL-route is meaningful and in line with traditional choice models.

Given that the proposed model is the first one applying deep learning for route choice in public transport, we see several future directions. First, there is the need of establishing in the literature deep learning models for route choice. Hence, in future works different models or neural network structures can be developed and compared with the proposed one. Similarly, novel datasets can be tested, with different external features or different network conditions (e.g. in case of an unreliable network). The interpretability of the model can be further studied, applying other methods of explainable artificial intelligence, like Layer-wise Relevance Propagation (Alwosheel et al. 2021), or evaluating the effect of features interaction. Finally, we can think that some of the concepts here introduced (CNN, explainability) can also inspire research in route choice in road transport or other modes, which shares the similar issue that alternatives are not uniquely labeled across observations, despite the setup and model specification is very different.

Appendix: PathSize factor

Since DL-route does not consider in input information on the path (i.e. the specific lines, streets or connections), it cannot capture correlations among the alternatives based on their paths. Therefore, the PathSize factor plays an important role in the model, capturing the correlations among the different paths in the choice set, as done in the PSL. Here, we study the inclusion and exclusion of the PathSize among the input features in both PSL (in its simplest form) and DL-route (Table 4).

In PSL, the PathSize is significant (t -test=2.49), and excluding it from the utility reduces slightly the performance: Log Likelihood from -0.78 to -0.79 ; prediction accuracy from 73

to 72%. In DL-route, the change in performance is marginal, depending also on the tuning of the model. These results show that for the analyzed dataset the PathSize is a less relevant feature compared to the others, as also shown in Marra and Corman (2020). The large Marginal Rate of Substitution of PathSize in Table 6 is due to its logarithmic nature (making PathSize smaller in absolute terms than other parameters).

The analyses in this work show a clear theoretical importance of considering a PathSize factor in DL-route, to capture correlations among overlapping alternatives. However, the dataset considered does not show significant differences in the performance. Therefore, for a future usage of DL-route, we recommend considering the PathSize, since it can capture correlations, which may be more relevant with a different dataset, or for private transport.

Acknowledgements The authors wish to thank all study participants

Author contributions All authors contributed to the study conception and design. Analyses and experiments were performed by ADM. All authors contributed to the writing of the manuscript. All authors read and approved the final manuscript.

Funding Open access funding provided by Swiss Federal Institute of Technology Zurich
Not applicable.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Ethics approval Approved by ETH Zurich Ethics Commission.

Consent to participate Informed consent was obtained from all participants included in the study.

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